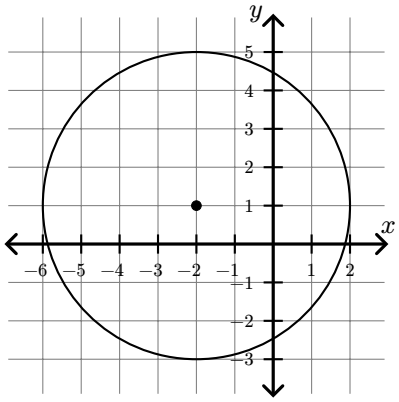
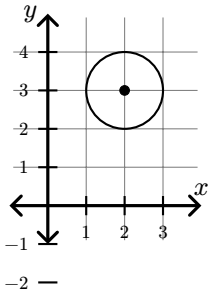
Section 6.3 — Circles — Answer Key

Math 103: College Algebra

Basic Equation of a Circle

On a separate sheet of paper, sketch each circles described below.

1. 

2. 

Give an equation of each circles described below.

3. $(x-3)^2+(y-5)^2=4$

4. $(x+5)^2+(y+2)^2=36$

Give an equation of the circle shown in each graph below.

5. $(x-5)^2+(y+4)^2=9$

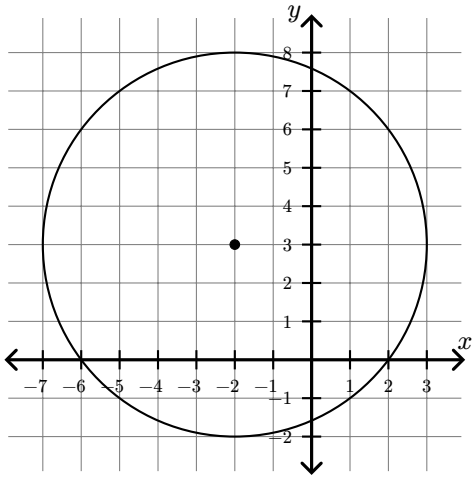
6. $(x+3)^2+(y-2)^2=25$

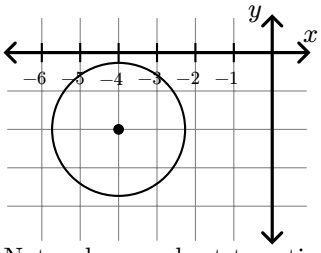
Give the center and radius of the circle described by each equation below.

7. Center: $(5,-2)$; Radius: 7

8. Center: $(-6,-1)$; Radius: $2\sqrt{10}$

On a separate sheet of graph paper, sketch the graph of each equation below.

9. 

10. 

Note: do your best to estimate the radius $\sqrt{3}\approx 1.732$

Writing Equations of Circles

Give the equation of each circle described below.

11. $(x-4)^2+(y-1)^2=41$

12. $(x-2)^2+(y+1)^2=50$

13. $(x+7)^2+(y-5)^2=13$

14. $(x+4)^2+(y+2)^2=20$

15. $(x+2)^2 + (y-4)^2 = 49$

16. $(x-3)^2 + (y+5)^2 = 9$

17. $(x+5)^2 + (y-1)^2 = 1$

18. $(x+4)^2 + (y+3)^2 = 34$

Give an equation of the circles graphed below.

19. $(x-2)^2 + (y+1)^2 = 5$

20. $(x-2)^2 + (y-2)^2 = 2$

Working with Standard Form

Give an equation of each circle below in standard form.

21. $x^2 + y^2 - 8x + 4y + 11 = 0$

22. $x^2 + y^2 + 4x + 10y + 19 = 0$

23. $x^2 + y^2 - 3x - 2y - 13 = 0$

24. $x^2 + y^2 - 4x - 2y - 4 = 0$

Rewrite each equation below in center-radius form. If the equation describes a true circle, give its center and radius. If it doesn't, state whether it is the equation of a degenerate circle (single point), or if the equation has no solution.

25. Center: $(2, 3)$; Radius: 2

26. Center: $(1, -3)$; Radius: 5

27. Center: $(1, -2)$; Radius: $\frac{3}{2}$

28. Center: $(0, -5)$; Radius: 4

29. Degenerate circle (single point $(3, 5)$)

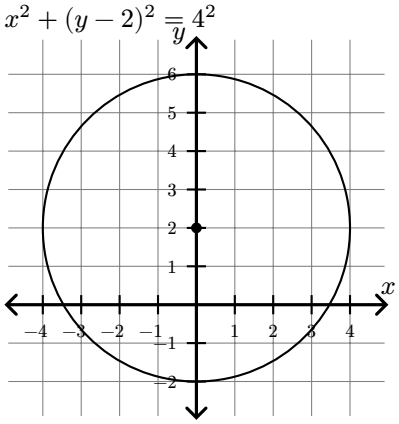
30. Center: $(-2, 4)$; Radius: 3

31. No solution

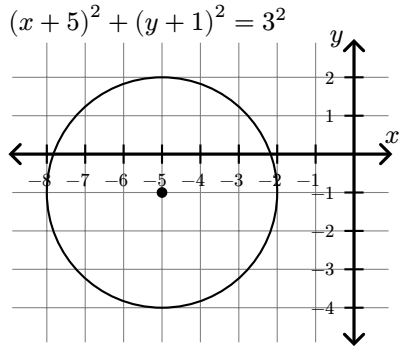
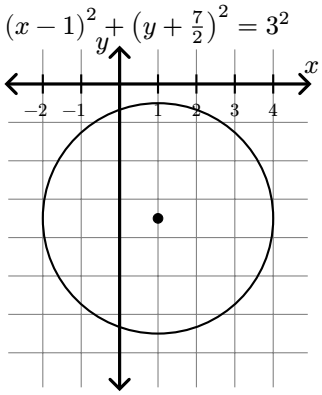
32. Degenerate circle (single point $(1, -5)$)

Rewrite each equation below in center-radius form. Then, graph the circle on a separate sheet of graph paper.

33. $x^2 + (y-2)^2 = 4^2$



34. $(x+5)^2 + (y+1)^2 = 3^2$



35. $(x-1)^2 + \left(y + \frac{7}{2}\right)^2 = 3^2$